

Q1 2021 (01 Sep Shift 2)

The function $f(x)$, that satisfies the condition

$f(x) = x + \int_0^{\pi/2} \sin x \cdot \cos y f(y) dy$, is:

(1) $x + \frac{2}{3}(\pi - 2) \sin x$

(2) $x + (\pi + 2) \sin x$

(3) $x + \frac{\pi}{2} \sin x$

(4) $x + (\pi - 2) \sin x$

Q2 2021 (31 Aug Shift 2)

If $[x]$ is the greatest integer $\leq x$, then

$\pi^2 \int_0^2 \left(\sin \frac{\pi x}{2} \right) (x - [x])^{[x]} dx$ is equal to :

(1) $2(\pi - 1)$

(2) $4(\pi - 1)$

(3) $4(\pi + 1)$

(4) $2(\pi + 1)$

Q3 2021 (31 Aug Shift 1)

Let $[t]$ denote the greatest integer $\leq t$. Then the value of $8 \cdot \int_{-\frac{1}{2}}^1 ([2x] + |x|) dx$ is .

Q4 2021 (31 Aug Shift 1)

Let f be a non-negative function in $[0, 1]$ and twice

differentiable in $(0, 1)$. If $\int_0^x \sqrt{1 - (f'(t))^2} dt = \int_0^x f(t) dt$,

$0 \leq x \leq 1$ and $f(0) = 0$, then $\lim_{x \rightarrow 0} \frac{1}{x^2} \int_0^x f(t) dt$:

(1) equals 0

(2) equals 1

(3) does not exist

(4) equals $\frac{1}{2}$

Q5 2021 (27 Aug Shift 2)

The value of the integral $\int_0^1 \frac{\sqrt{x} dx}{(1+x)(1+3x)(3+x)}$ is:

- (1) $\frac{\pi}{8} \left(1 - \frac{\sqrt{3}}{2}\right)$
- (2) $\frac{\pi}{4} \left(1 - \frac{\sqrt{3}}{6}\right)$
- (3) $\frac{\pi}{8} \left(1 - \frac{\sqrt{3}}{6}\right)$
- (4) $\frac{\pi}{4} \left(1 - \frac{\sqrt{3}}{2}\right)$

Q6 2021 (27 Aug Shift 1)

$\int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e (x^2 - 44x + 484)} dx$ is equal to:

- (1) 6
- (2) 8
- (3) 5
- (4) 10

Q7 2021 (27 Aug Shift 1)

If $U_n = \left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2^2}{n^2}\right)^2 \dots \left(1 + \frac{n^2}{n^2}\right)^n$, then $\lim_{n \rightarrow \infty} (U_n)^{\frac{-4}{n^2}}$ is equal to :

- (1) $\frac{e^2}{16}$
- (2) $\frac{4}{e}$
- (3) $\frac{16}{e^2}$
- (4) $\frac{4}{e^2}$

Q8 2021 (26 Aug Shift 2)

The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1 + \sin^2 x}{1 + \pi \sin x} \right) dx$ is

- (1) $\frac{\pi}{2}$
- (2) $\frac{5\pi}{4}$
- (3) $\frac{3\pi}{4}$

(4) $\frac{3\pi}{2}$

Q9 2021 (26 Aug Shift 2)

If the value of the integral $\int_0^5 \frac{x+[x]}{e^{x-[x]}} dx = \alpha e^{-1} + \beta$,

where $\alpha, \beta \in \mathbf{R}$, $5\alpha + 6\beta = 0$, and $[x]$ denotes the

greatest integer less than or equal to x ; then the value of $(\alpha + \beta)^2$ is equal to :

(1) 100

(2) 25

(3) 16

(4) 36

Q10 2021 (26 Aug Shift 1)

The value of $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{2n-1} \frac{n^2}{n^2 + 4r^2}$ is:

(1) $\frac{1}{2} \tan^{-1}(2)$

(2) $\frac{1}{2} \tan^{-1}(4)$

(3) $\tan^{-1}(4)$

(4) $\frac{1}{4} \tan^{-1}(4)$

Q11 2021 (26 Aug Shift 1)

The value of $\int_{-1/\sqrt{2}}^{1/\sqrt{2}} \left(\left(\frac{x+1}{x-1} \right)^2 + \left(\frac{x-1}{x+1} \right)^2 - 2 \right)^{1/2} dx$ is:

(1) $\log_e 4$

(2) $\log_e 16$

(3) $2 \log_e 16$

(4) $4 \log_e (3 + 2\sqrt{2})$

Answer Key

Q1 (4)

Q2 (2)

Q3 (5)

Q4 (4)

Q5 (1)

Q6 (3)

Q7 (1)

Q8 (3)

Q9 (2)

Q10 (2)

Q11 (2)

Q1 (4)

$$f(x) = x + \int_0^{\pi/2} \sin x \cos y f(y) dy$$

$$f(x) = x + \sin x \underbrace{\int_0^{\pi/2} \cos y f(y) dy}_K$$

$$\Rightarrow f(x) = x + K \sin x$$

$$\Rightarrow f(y) = y + K \sin y$$

$$\text{Now } K = \int_0^{\pi/2} \cos y (y + K \sin y) dy$$

$$K = \int_0^{\pi/2} y \cos y dy + \int_0^{\pi/2} \cos y \sin y y dy$$

$$K = (y \sin y)_0^{\pi/2} - \int_0^{\pi/2} \sin y dy + K \int_0^1 t dt$$

$$\Rightarrow K = \frac{\pi}{2} - 1 + K \left(\frac{1}{2} \right)$$

$$\Rightarrow K = \pi - 2$$

$$\text{So } f(x) = x + (\pi - 2) \sin x$$

Option (4)

Q2 (2)

$$\pi^2 \left[\int_0^1 \sin \frac{\pi x}{2} dx + \int_1^2 \sin \frac{\pi x}{2} (x-1) dx \right]$$

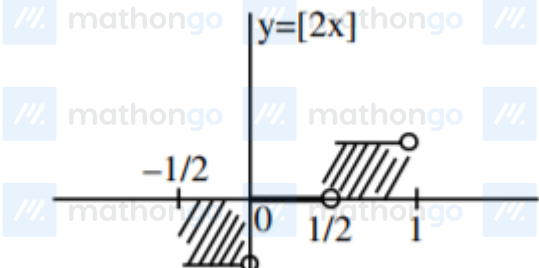
$$= \pi^2 \left[-\frac{2}{\pi} \left(\cos \frac{\pi x}{2} \right) + \left((x-1) \left(-\frac{2}{\pi} \cos \frac{\pi x}{2} \right) \right)_1^2 - \int_1^2 -\frac{2}{\pi} \cos \frac{\pi x}{2} dx \right]$$

$$= \pi^2 \left[0 + \frac{2}{\pi} + \frac{2}{\pi} + \frac{2}{\pi} \cdot \frac{2}{\pi} \left(\sin \frac{\pi x}{2} \right)_1^2 \right]$$

$$= 4\pi - 4 = 4(\pi - 1)$$

Q3 (5)

$$I = \int_{-1/2}^1 ([2x] + |x|) dx$$



$$\begin{aligned}
 &= \int_{-1/2}^1 [2x] dx + \int_{-1/2}^1 |x| dx \\
 &= 0 + \int_{-1/2}^0 (-x) dx + \int_0^1 x dx \\
 &= \left(-\frac{x^2}{2}\right)_{-1/2}^0 + \left(\frac{x^2}{2}\right)_0^1 \\
 &= \left(0 + \frac{1}{8}\right) + \frac{1}{2} \\
 &= \frac{5}{8} \\
 8I &= 5
 \end{aligned}$$

Q4 (4)

$$\int_0^x \sqrt{1 - (f'(t))^2} dt = \int_0^x f(t) dt \quad 0 \leq x \leq 1$$

differentiating both the sides

$$\begin{aligned}
 \sqrt{1 - (f'(x))^2} &= f(x) \\
 \Rightarrow 1 - (f'(x))^2 &= f^2(x)
 \end{aligned}$$

$$\frac{f'(x)}{\sqrt{1 - f^2(x)}} = 1$$

$$\sin^{-1} f(x) = x + C$$

$$\because f(0) = 0 \Rightarrow C = 0 \Rightarrow f(x) = \sin x$$

$$\text{Now } \lim_{x \rightarrow 0} \frac{\int_0^x \sin t dt}{x^2} \left(\frac{0}{0} \right) = \frac{1}{2}$$

Q5 (1)

$$I = \int_0^1 \frac{\sqrt{x}}{(1+x)(1+3x)(3+x)} dx$$

$$\text{Let } x = t^2 \Rightarrow dx = 2t \cdot dt$$

$$I = \int_0^1 \frac{t(2t)}{(t^2+1)(1+3t^2)(3+t^2)} dt$$

$$I = \int_0^1 \frac{(3t^2+1) - (t^2+1)}{(3t^2+1)(t^2+1)(3+t^2)} dt$$

$$\begin{aligned}
 I &= \int_0^1 \frac{dt}{(t^2+1)(3+t^2)} - \int_0^1 \frac{dt}{(1+3t^2)(3+t^2)} \\
 &= \frac{1}{2} \int_0^1 \frac{(3+t^2) - (t^2+1)}{(t^2+1)(3+t^2)} dt + \frac{1}{8} \int_0^1 \frac{(1+3t^2) - 3(3+t^2)}{(1+3t^2)(3+t^2)} dt \\
 &= \frac{1}{2} \int_0^1 \frac{dt}{1+t^2} - \frac{1}{2} \int_0^1 \frac{dt}{t^2+3} + \frac{1}{8} \int_0^1 \frac{dt}{t^2+3} - \frac{3}{8} \int_0^1 \frac{dt}{(1+3t^2)} \\
 &= \frac{1}{2} \int_0^1 \frac{dt}{t^2+1} - \frac{3}{8} \int_0^1 \frac{dt}{t^2+3} - \frac{3}{8} \int_0^1 \frac{dt}{1+3t^2}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} (\tan^{-1}(t))_0^1 - \frac{3}{8\sqrt{3}} \left(\tan^{-1}\left(\frac{t}{\sqrt{3}}\right) \right)_0^1 \\
 &\quad - \frac{3}{8\sqrt{3}} \left(\tan^{-1}(\sqrt{3}t) \right)_0^1 \\
 &= \frac{1}{2} \left(\frac{\pi}{4} \right) - \frac{\sqrt{3}}{8} \left(\frac{\pi}{6} \right) - \frac{\sqrt{3}}{8} \left(\frac{\pi}{3} \right) \\
 &= \frac{\pi}{8} - \frac{\sqrt{3}}{16} \pi \\
 &= \frac{\pi}{8} \left(1 - \frac{\sqrt{3}}{2} \right)
 \end{aligned}$$

Q6 (3)

$$\begin{aligned}
 \text{Let } I &= \int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e (x^2 - 44x + 484)} dx \\
 I &= \int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e (x - 22)^2} dx \dots (1)
 \end{aligned}$$

We know

$$\int_a^b f(x) dx = \int_a^b f(a + b - x) dx \text{ (king)}$$

$$\begin{aligned}
 \text{So } I &= \int_6^{16} \frac{\log_e (22 - x)^2}{\log_e (22 - x)^2 + \log_e (22 - (22 - x))^2} dx \\
 I &= \int_0^{16} \frac{\log_e (22 - x)^2}{\log_e x^2 + \log_e (22 - x)^2} dx \dots (2)
 \end{aligned}$$

$$(1) + (2)$$

$$\begin{aligned}
 2I &= \int_6^{16} 1 \cdot dx = 10 \\
 I &= 5
 \end{aligned}$$

Q7 (1)

$$U_n = \prod_{r=1}^n \left(1 + \frac{r^2}{n^2} \right)^r$$

$$L = \lim_{n \rightarrow \infty} (U_n)^{-4/n^2}$$

$$\begin{aligned}
 \log L &= \lim_{n \rightarrow \infty} \frac{-4}{n^2} \sum_{r=1}^n \log \left(1 + \frac{r^2}{n^2} \right)^r \\
 \Rightarrow \log L &= \lim_{n \rightarrow \infty} \sum_{r=1}^n -\frac{4r}{n} \cdot \frac{1}{n} \log \left(1 + \frac{r^2}{n^2} \right)
 \end{aligned}$$

$$\Rightarrow \log L \Rightarrow -4 \int_0^1 x \log(1 + x^2) dx$$

$$\begin{aligned}
 \text{put } 1 + x^2 &= t \\
 \text{Now, } 2x dx &= dt
 \end{aligned}$$

$$= -2 \int_1^2 \log(t) dt = -2 [t \log t - t]_1^2$$

$$\Rightarrow \log L = -2(2 \log 2 - 1)$$

$$\therefore L = e^{-2(2 \log 2 - 1)}$$

$$= e^{-2\left(\log\left(\frac{4}{e}\right)\right)}$$

$$= e^{\log\left(\frac{4}{e}\right)^{-2}}$$

$$= \left(\frac{e}{4}\right)^2 = \frac{e^2}{16}$$

Q8 (3)

$$I = \int_0^{\pi/2} \frac{(1+\sin^2 x)}{(1+\pi^{\sin x})} + \frac{\pi^{\sin x}(1+\sin^2 x)}{(1+\pi^{\sin x})} dx$$

$$I = \int_0^{\pi/2} (1 + \sin^2 x) dx$$

$$I = \frac{\pi}{2} + \frac{\pi}{2} \cdot \frac{1}{2} = \frac{3\pi}{4}$$

Q9 (2)

$$I = \int_0^5 \frac{x+[x]}{e^{x-[x]}} dx$$

$$\int_0^1 \frac{x}{e^x} dx + \int_1^2 \frac{x+1}{e^{x-1}} dx + \int_2^3 \frac{x+2}{e^{x-2}} dx + \dots + \int_4^5 \frac{x+4}{e^{x-4}} dx$$

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$$x = t + 1 \quad x = z + 2 \quad x = y + 4$$

$$\int_0^1 \frac{t+2}{e^t} dt + \int_0^1 \frac{z+4}{e^z} dz + \dots + \int_0^1 \frac{y+8}{e^y} dy$$

$$\Rightarrow \int_0^5 \frac{5x+20}{e^x} dx = 5 \int_0^1 \frac{x+4}{e^x} dx$$

$$\Rightarrow 5 \int_0^1 (x+4)e^{-x} dx$$

$$\Rightarrow 5e^{-x}(-x-5)|_0^1 \Rightarrow -\frac{30}{e} + 25$$

$$\alpha = -30$$

$$\beta = 25 \Rightarrow 5\alpha + 6\beta = 0$$

$$(\alpha + \beta)^2 = 5^2 = 25$$

Q10 (2)

$$L = \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \sum_{r=0}^{2n-1} \frac{1}{1+4\left(\frac{r}{n}\right)^2}$$

$$\Rightarrow L = \int_0^2 \frac{1}{1+4x^2} dx$$

$$\Rightarrow L = \left. \frac{1}{2} \tan^{-1}(2x) \right|_0^2 \Rightarrow L = \frac{1}{2} \tan^{-1} 4$$

Q11 (2)

$$I = \int_{-1/\sqrt{2}}^{1/\sqrt{2}} \left(\left(\frac{x+1}{x-1} - \frac{x-1}{x+1} \right)^2 \right)^{1/2} dx$$

$$I = \int_{-1/\sqrt{2}}^{1/\sqrt{2}} \left| \frac{4x}{x^2-1} \right| dx \Rightarrow I = 2.4 \int_0^{1/\sqrt{2}} \left| \frac{x}{x^2-1} \right| dx$$

$$\Rightarrow I = -4 \int_0^{1/\sqrt{2}} \frac{2x}{x^2-1} dx \Rightarrow I = -4 \ln |x^2-1|_0^{1/\sqrt{2}}$$

$$\Rightarrow I = 4 \ln 2 \Rightarrow I = \ln 16$$